Noesis: Selective Memory Injection for Cross-Tool AI Conversation Consistency

Peizheng Du
Independent Researcher
peterdu0628@gmail.com

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Abstract

Users converse with multiple AI assistants daily, yet each tool's memory is isolated: a stance expressed in one tool is unknown to the others, and each conversation starts from scratch. Existing approaches either store flat facts without a notion of confidence, or inject all retrieved context regardless of relevance, causing token costs to explode with memory size. I present **Noesis**, a local-first memory layer that (1) extracts typed *thoughts* from conversations across tools into a shared per-user store, (2) gates injection behind a confidence lifecycle in which thoughts mature from *tentative* to *settled* based on repetition, assertion strength, cross-tool corroboration, and contradiction signals, and (3) caps injected context with a token budget. Across three models (Gemini Flash-Lite, gemma3:4b, qwen2.5:3b) on 110 paired questions each, memory injection raises answer accuracy from 41.8% to 85.5%, 34.5% to 82.7%, and 37.3% to 62.7% respectively (all $p < 0.001$, McNemar's test). At 500 stored memories, Noesis keeps 9 of 10 targets at rank 1 with about 207 injected tokens, 35 times fewer than injecting every memory (MRR 0.90 vs. 0.002); a standard top-5 RAG ranks as well with fewer tokens, so Noesis's distinct gain is filtering: where ungated retrieval surfaced unverified or planted statements 19 and 13 times in two tests, Noesis surfaced none. On strictly cross-tool queries, Noesis retrieves 37% where per-tool isolation is structurally bound to 0%. I release the system, experiment harness, and all raw results.

Keywords: large language models; LLM memory; retrieval-augmented generation; personalization; cross-tool memory; confidence gating; local-first software

1 Introduction

Heavy users of AI assistants switch between ChatGPT, Claude, Gemini, and local models throughout the day. Each tool maintains its own isolated memory: a database preference carefully explained to Claude must be re-explained to GPT; a design decision reached with Gemini is invisible to Claude. The reasoning a user performs across tools, such as their positions, preferences, and open questions, evaporates at every tool boundary.

Two families of systems address parts of this problem, each with a critical gap. *Fact stores* such as Mem0 [2] persist extracted facts but treat all memories equally: a firmly held professional identity and a passing mention of a tool have the same standing, and there is no mechanism for a memory to earn trust over time. *Retrieval-augmented generation* (RAG) [5] retrieves relevant context for each query, but naive variants inject everything retrieved; as personal memory grows into the hundreds of items, injected context grows linearly, burying the relevant signal and inflating cost (I measure 7,192 tokens at n=500).

Noesis is built on a different premise: *what* gets injected matters as much as *whether* anything is injected. Thoughts should earn their way into the context the way beliefs earn their way into a mind: through repetition, consistency, and survival against contradiction.

This paper makes the following contributions:

1. **A confidence lifecycle for conversational memory** (§3). Thoughts are typed (identity, preference, position, question, synthesis, event, contradiction) and mature tentative→provisional→settled via a four-signal scorer, with type-specific time decay and cluster-aware supersession that reduced stale-stance leakage from 93% to 7% (§3.3).

2. **Gated injection that scales** (§5.2–§5.3). At 500 memories Noesis injects ∼207 tokens (35 times fewer than injecting everything) while holding MRR at 0.90; a standard top-5 RAG matches that ranking at lower cost, and Noesis's gate is what keeps unverified and planted statements out (0 vs. 19 and 0 vs. 13).

3. **Quantified cross-tool memory sharing** (§5.4). With a shared per-user store, Noesis retrieves 37% of strictly cross-tool queries versus 0% for per-tool isolation (50% vs. 7.5% over all 40 scenarios)—a gap that is architecturally impossible for isolated systems to close.

4. **An A/B result across three model tiers** (§5.1). On identical n=110 question sets, memory injection significantly improves answer accuracy for a cloud model (Gemini Flash-Lite, +43.6pp), a mid-tier local model (gemma3:4b, +48.2pp), and a small local model (qwen2.5:3b, +25.5pp); all $p < 0.001$.

The system is local-first: embeddings and retrieval run on-device (sqlite-vec + all-MiniLM-L6-v2), and only optional LLM-based extraction calls a cloud API. All code, the experiment harness, raw results, and a verification script that recomputes every reported number from raw JSON are released.[1]

2 Related Work

Memory for LLM agents. MemGPT [9] treats the LLM as an operating system paging memory between contexts; Letta extends this with archival storage. Mem0 [2] extracts and updates atomic facts with an LLM pipeline, reporting gains over full-history baselines; A-MEM [14] links memories into an evolving Zettelkasten-style network, and Zep [11] stores them in a temporal knowledge graph. Generative Agents [10] rank memories by recency, importance, and relevance, and MemoryBank [16] lets them fade along an Ebbinghaus forgetting curve, the closest precedents for Noesis's type-specific decay. OpenMemory MCP [8] already shares one local store across MCP clients such as Cursor and Claude Desktop. Noesis differs in (a) its lifecycle, where a memory must earn injection rights before it reaches the prompt, and (b) its entry points beyond MCP: an OpenAI-compatible proxy and a browser extension make clients outside MCP first-class participants. Benchmarks such as LoCoMo [6] and LongMemEval [13] offer standard long-term-memory evaluations that this paper does not yet use.

Retrieval-augmented generation. RAG [5] and successors retrieve per-query context. Self-RAG [1] and FLARE [4] improve retrieval quality or adaptivity, but address corpus QA, not a user's evolving personal memory. My scale experiment (§5.3) shows why naive inject-everything

[1] https://github.com/AngryKiwiWithLegs/Noesis

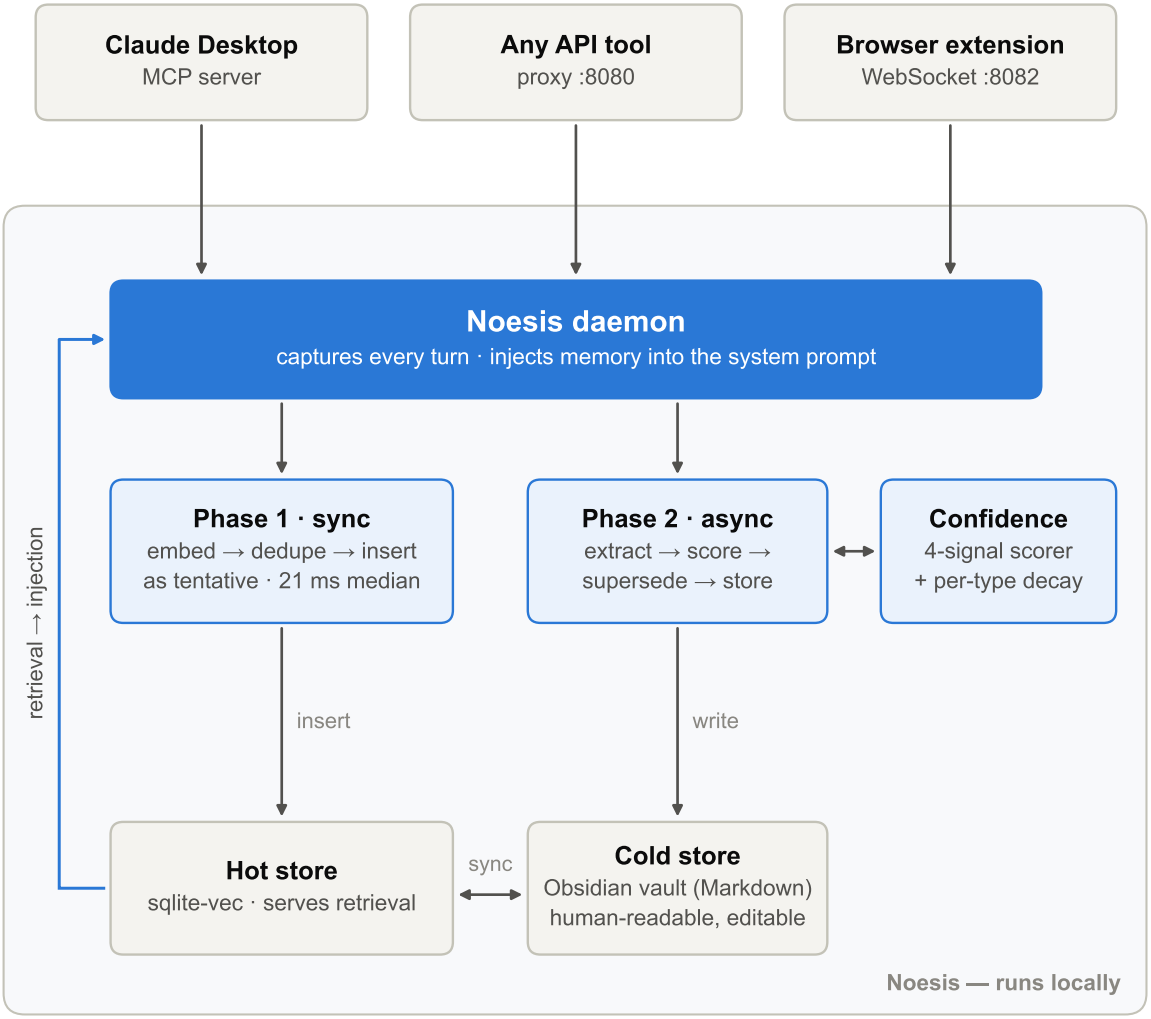


Figure 1: Noesis system architecture. Tools connect through an MCP server, an OpenAI-compatible proxy, or a WebSocket; Phase 1 stores the utterance synchronously (median 21 ms) and Phase 2 asynchronously extracts, scores, and links the thought. Hot store (sqlite-vec) serves retrieval; cold store (Obsidian vault) keeps memory human-readable and editable.

retrieval fails for personal memory: cost grows linearly and ranking collapses. Noesis pairs hybrid retrieval (BM25 + vector + RRF, with time-decay weighting) with a hard token budget and a confidence gate.

Personalization. PALR [15] and related work personalize outputs from user interaction history, typically for recommendation-style tasks. Chat-tool vendor memory features (e.g., ChatGPT memory) are per-product and closed. Noesis's cross-tool experiment (§5.4) demonstrates a structural advantage of user-scoped rather than tool-scoped memory.

Visualization of embedding spaces. t-SNE [12] and UMAP [7] project high-dimensional embeddings for inspection; force-directed layouts [3] model attraction/repulsion in 2D. My star map (§6) extends force-directed layout to 3D and adds weak axis springs, so semantic clusters still form freely while time reads left to right and confidence bottom to top.

3 System Design

Noesis sits between AI tools and model providers (Figure 1). API clients point at a local OpenAI-compatible proxy, which injects memory into the system prompt, forwards the request to the configured provider (cloud or local), and captures the conversation for memory formation; Claude Desktop connects through an MCP server and the browser extension through a WebSocket.

3.1 Thought taxonomy and two-phase capture

Conversational memory is typed into seven categories with distinct persistence semantics (Table 1). An *identity* thought ("I am a backend engineer") should outlive an *event* ("I tried Bun yesterday"), and the taxonomy encodes this as type-specific half-lives.

Capture is two-phase. *Phase 1* is synchronous: embed, deduplicate by hash, insert as *tentative* (median 21 ms, p99 684 ms over 100 adds on an x86-64 Mac). *Phase 2* runs in the background, so the conversation never waits for it; building the injected context adds a median of 21 ms (p99 55 ms) per request. In *Phase 2*, an extractor (a small cloud LLM, or a deterministic fallback) classifies the thought; a confidence scorer updates its status; a supersession check retires contradicted predecessors; the cold store (Markdown + YAML frontmatter in an Obsidian vault) is updated for human readability and editability.

Table 1: Thought taxonomy with type-specific half-lives and injection priority.

Type	**Definition**	**Half-life (days)**	**Priority**
identity	Who the user is	∞	highest (always)
preference	Explicit preference	90	high
position	Stance on a topic	90	high
question	Open question under exploration	30	medium
synthesis	Insight from merged ideas	180	medium
event	Happened / decided	7	low (recent)
contradiction	Self-contradictory judgment	∞	flagged

3.2 Confidence lifecycle

Every thought begins *tentative*: stored, but never injected and invisible to retrieval—the "capture threshold low, injection threshold high" principle. Four signals promote a thought: *repetition* (the same stance recurs across sessions), *assertion strength* ("I've decided" outranks "maybe"), *cross-tool consistency* (distinct phrasings of a stance from different tools), and *absence of contradiction*. The scorer combines them as $0.35\times$repetition$+0.25\times$assertion$+0.25\times$no-contradiction$+0.15\times$cross-tool, each scored from 0 to 1; a thought becomes provisional at 0.45 and settled at 0.75. A single firmly stated thought with no conflict ("I prefer X", assertion 0.8) therefore scores exactly 0.45 and becomes injectable, while settling always requires repetition: at least three similar earlier thoughts with cross-tool corroboration, or six without. *Provisional* thoughts inject when topic-relevant; *settled* thoughts always inject. Confidence evolves incrementally: reinforcement $+0.10\,(1 - C)$, conflict -0.20, and time decay $C \leftarrow C\,e^{-0.693\,\Delta t/t_{1/2}}$ with $t_{1/2}$ set by type (Table 1).

3.3 Cluster-aware supersession

When a user changes their mind ("I moved from MySQL to PostgreSQL"), the old stance should retire. Embedding similarity is the wrong signal here: contradictory stances mention different entities and therefore have *low* similarity. Noesis detects mind-changes by requiring jointly (1) a replacement verb ("switched to", "migrated to", …, EN+ZH), (2) the same domain via a keyword map (both stances are about *databases*), (3) stance-like type, and (4) injectable status of the new thought. Matched predecessors are marked *superseded*—excluded by every retrieval path—and linked via `superseded_by`. In my evaluation (the same 30 scenarios, run once before and once after the fix) this reduced stale-stance leakage from 93% to 7% (Table 2).

Table 2: Supersession fix: stale-stance leakage before/after (n=30 scenarios).

Metric	**Before**	**After**	**Δ**
Old stance leaked into context	28/30 (93%)	2/30 (7%)	−87pp
New stance surfaced	12/30 (40%)	28/30 (93%)	+53pp
Ideal (new shown, old retired)	1/30 (3%)	27/30 (90%)	+87pp

3.4 Selective retrieval and injection

At query time, three channels run in parallel (Figure 2): BM25 for exact terms, dense retrieval for paraphrase, and two lightweight signals (recency, core-facts). Channels fuse via reciprocal-rank fusion with time-decay weighting. A confidence gate drops everything below *provisional*, and a hard token budget (1,200 tokens) truncates the remainder. The result is injected into the system prompt as compact structured lines.

4 Evaluation Setup

I evaluate four research questions:

- **RQ1**: Does memory injection improve answer accuracy? (A/B, end-to-end, §5.1)
- **RQ2**: Which components contribute? (ablation, retrieval-layer, §5.2)

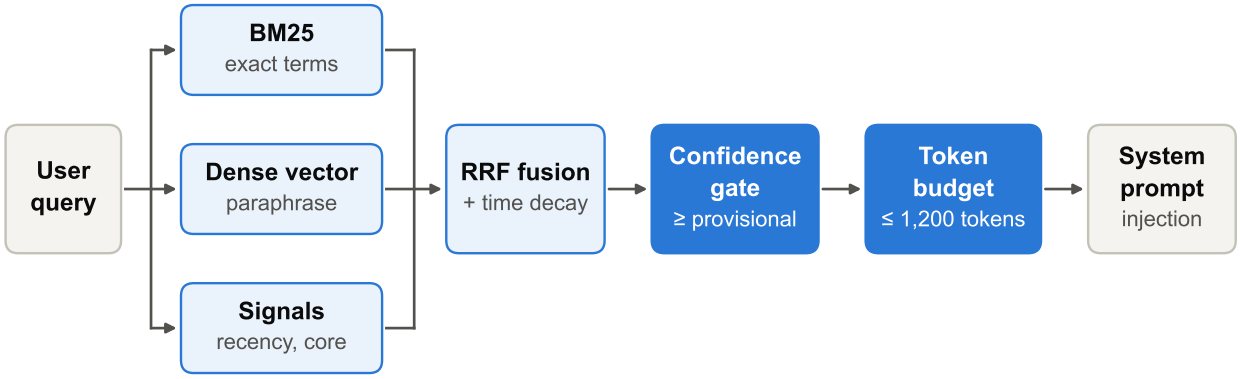


Figure 2: Retrieval and injection pipeline. Three channels fuse via RRF; a confidence gate removes non-injectable thoughts; a token budget caps injected context.

- **RQ3**: How do cost and ranking scale with memory size? (§5.3)
- **RQ4**: Does cross-tool sharing work? (§5.4)

Models. Gemini Flash-Lite (cloud; `gemini-flash-lite-latest`, queried 30 June 2026), gemma3:4b and qwen2.5:3b (local via Ollama), spanning capability tiers.

A/B design. 30 synthetic English user profiles, each with 5 build statements (an identity statement plus preferences, positions, and decisions) followed by 5 test questions whose expected keyword is known from the build statements. Each build statement is stated once and firmly ("I prefer…", "I decided…"), which is enough to reach provisional (§3.2). For each question the model answers twice: *with memory* (via the Noesis proxy, which injects context) and *without memory* (direct to the provider). All models are evaluated on the identical first 110 questions (22 profiles). Scoring is objective keyword-hit; paired outcomes are tested with McNemar's χ^2.

Retrieval-layer experiments. Ablation, scale, and cross-tool experiments run against the real pipeline without LLM calls, measuring Recall@5, MRR, the recall and size of the full injected context, tentative leakage, and injected tokens.

Reproducibility. Every experimental number in this paper is recomputed from the raw result files by a released script (`experiments/verify_paper.py`), which also reads the lifecycle constants from the source. The ablation and scale experiments were re-run in October 2026 with the June software versions; they reproduced the June numbers exactly before the corrected leak metric and the top-5 baseline were added. The ablation was later corrected further (§9).

5 Results

5.1 RQ1: Memory injection improves accuracy across model tiers

Table 3 and Figure 3 show the headline result. Three findings stand out. First, *injection helps everywhere*: all three models improve significantly, and memory *hurt* in only 10 of 330 paired answers (3.0%). Second, *memory levels the field between local and cloud*: with injection, gemma3:4b

Table 3: A/B results on identical n=110 questions. All $p < 0.001$ (McNemar, df=1).

Model	n	**With**	**Without**	Δpp	χ^2
Gemini Flash-Lite	110	85.5%	41.8%	**+43.6**	42.48
gemma3:4b	110	82.7%	34.5%	**+48.2**	51.02
qwen2.5:3b	110	62.7%	37.3%	**+25.5**	16.57

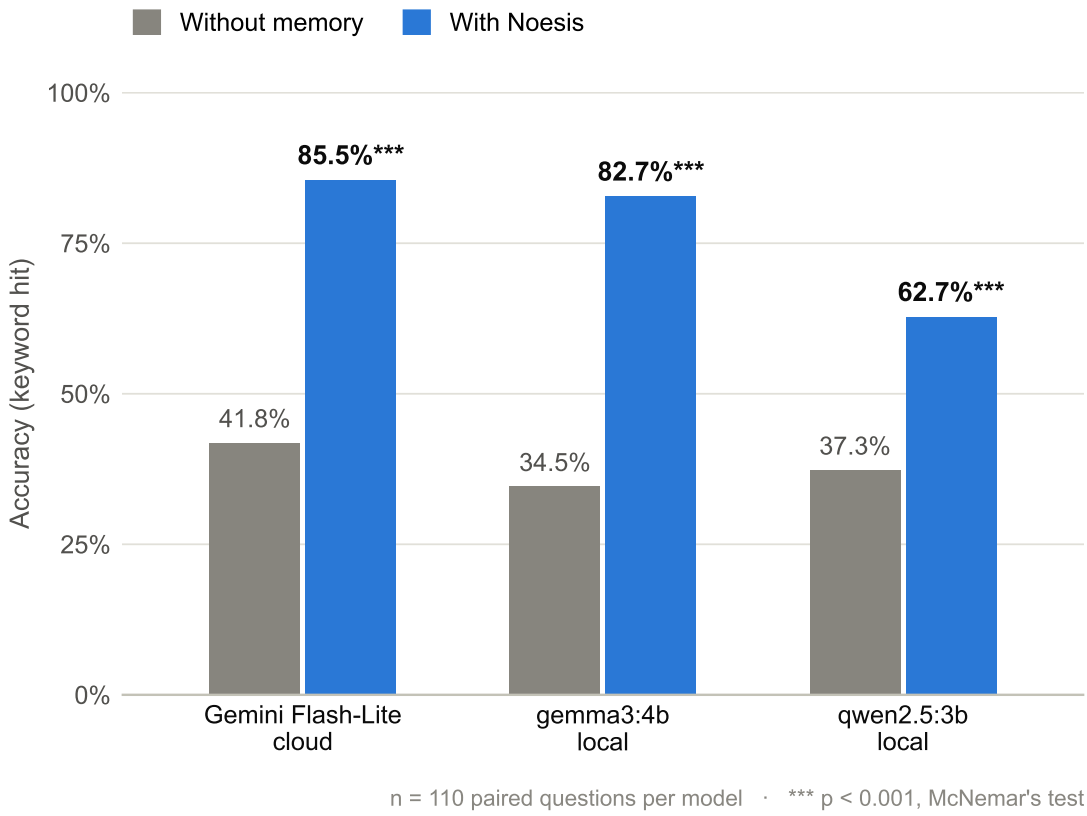


Figure 3: A/B accuracy across models (n=110 each). All improvements significant at $p < 0.001$.

(4B parameters, local) reaches 82.7%, nearly matching Gemini Flash-Lite's 85.5%, while its no-memory baseline (34.5%) trails Gemini's 41.8%. Third, *small models appear comprehension-capped*: qwen2.5:3b goes through the same retrieval pipeline but reaches only 62.7%, which points to a model-capability ceiling rather than a retrieval failure; the saved runs do not record the injected context, so this is not yet confirmed.

Accuracy alone does not separate retrieval strategies at this scale. In a separate end-to-end run on 50 questions (10 profiles, Gemini Flash-Lite), Noesis, injecting every stored memory, and injecting the five most recent memories scored 41, 40, and 41 of 50, against 22 of 50 with no memory. With only 3–9 stored memories per profile in that run, all three strategies inject nearly the same context; they diverge on filtering and cost as the store grows (§5.2–§5.3).

5.2 RQ2: Semantic retrieval drives recall; gating keeps unverified thoughts out

Table 4: Ablation and baselines (50 queries over 28 memories, 5 of them tentative; retrieval-layer, 2,000-token budget). A retrieved memory counts only if it is the ground-truth memory itself. Recall@5 and MRR use the first five memories in injection order; context recall, items, and leak cover everything the configuration injects, and leak counts tentative memories among those slots.

Config	Recall@5	MRR	Context recall	Items	Leak
Noesis (full)	**99%**	**1.000**	**100%**	**10.9**	**0/546**
w/o core-fact signal	99%	1.000	100%	9.7	0/487
w/o recency signal	99%	1.000	100%	9.7	0/484
w/o confidence gating	99%	1.000	100%	11.5	**73/573**
w/o semantic retrieval	24%	0.105	24%	5.0	0/250
Top-5 RAG	99%	1.000	99%	5.0	19/250
Recent-window	17%	0.084	17%	5.0	50/250
Random	12%	0.063	12%	5.0	51/250

Each Noesis variant runs the real context builder with exactly one component switched off, so signal order, deduplication, and the token budget stay the same; the baselines draw from every stored memory, tentative ones included. Status and confidence are assigned when the test store is built, so this suite tests whether retrieval respects those labels, not whether the lifecycle derives them from conversation. With 28 memories the semantic channel is dense retrieval alone (BM25 fusion starts at 300 memories).

Removing semantic retrieval collapses recall (99%→24%), confirming it as the main source of recall (Table 4). Removing the confidence gate leaves recall unchanged but lets tentative thoughts into 73 of the 573 injected slots, where the full system injects none—gating is what keeps *unverified* material out of the prompt. Ungated top-5 retrieval ranks as well as Noesis here (Recall@5 99%, MRR 1.000), as it does at scale (§5.3), but injects a tentative thought in 19 of its 250 slots. A separate poisoning test gives the same picture: with 5 settled facts and 8 hedged statements that contradict them, ungated top-5 retrieval returned a hedged statement in 13 of 25 slots across 5 queries, and Noesis in none. Removing the core-fact or recency signal changes nothing measurable: each adds about one memory per query (10.9 vs. 9.7 items) without changing recall. The builder adds these signals after the eight semantic results, so they never reach the first five slots; whether they help on queries the semantic channel misses needs a dedicated query set.

5.3 RQ3: Retrieval keeps cost flat as memory grows

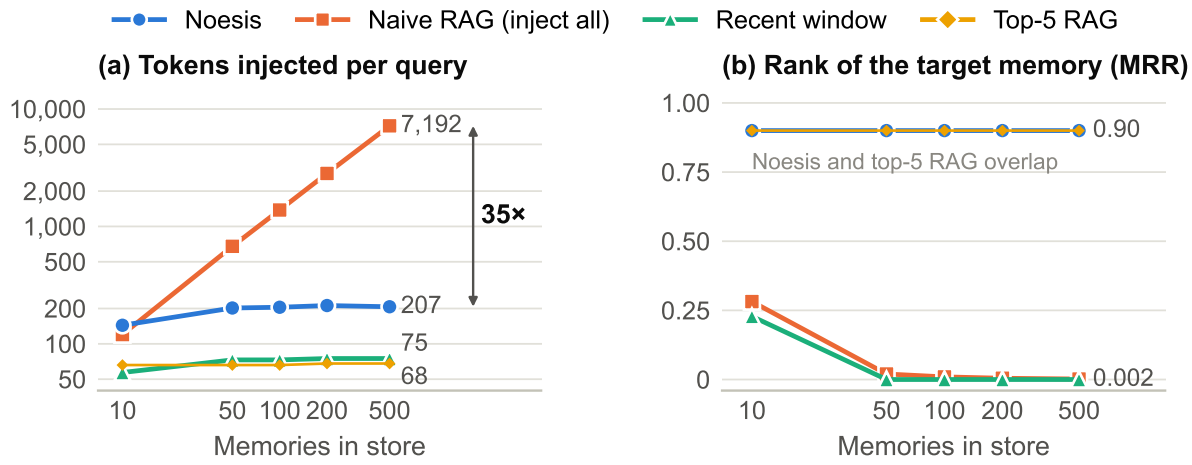


Figure 4: Scale behavior from 10 to 500 stored memories. (a) Noesis's injected tokens stay flat (∼207) while naive RAG (inject all) grows to 7,192 (35 times); top-5 RAG stays at 66–68. (b) Noesis and top-5 RAG hold MRR at 0.90; naive RAG collapses to 0.002 as targets are buried; recent-window loses recall entirely past n=10.

Figure 4 sweeps memory size from 10 to 500 with 10 known targets embedded among fillers. Noesis's injected cost is flat (144→207 tokens) because retrieval returns a small, confidence-gated set; the 1,200-token budget is never reached. MRR holds at 0.90: 9 of the 10 targets sit at rank 1, and the tenth is never retrieved. Naive RAG keeps recall only by injecting everything: 7,192 tokens at n=500, with the target buried near rank 500 (MRR 0.002). Recent-window recall collapses to 0% beyond n=10: the window forgets everything older. A standard top-5 RAG, which embeds the query and injects its five nearest memories, does as well as Noesis on this benchmark: the same 9 of 10 targets at rank 1, with 66–68 tokens per query. Flat cost and good ranking therefore come from retrieval itself rather than from anything specific to Noesis; what Noesis adds is the gate (§5.2), which top-5 RAG lacks. The methodological takeaway: *recall alone is misleading*; rank and token cost must be reported together.

5.4 RQ4: Cross-tool sharing vs. structural isolation

Of 40 scenarios, 27 state a stance in one tool and ask about it from another; in 7, the asking tool was one of two tools that stated it; and in 6, a third tool asks after two others stated it (Figure 5). On the 27 strictly cross-tool queries, Noesis retrieves 37% (10/27) and the isolated baseline 0%—not because it ranks poorly, but because the memory is not in its partition, so that 0% holds by construction. The 7 scenarios where the asking tool did see the stance are a fairer comparison: isolated memory retrieves 3 of 7 and Noesis 6 of 7. Over all 40 scenarios, Noesis retrieves 50% and the isolated baseline 7.5%. This is the structural argument: per-tool memory systems cannot share by design, and no amount of per-tool tuning closes the gap. Noesis's misses trace to retrieval precision (the memory exists but ranks below top-k), not to sharing.

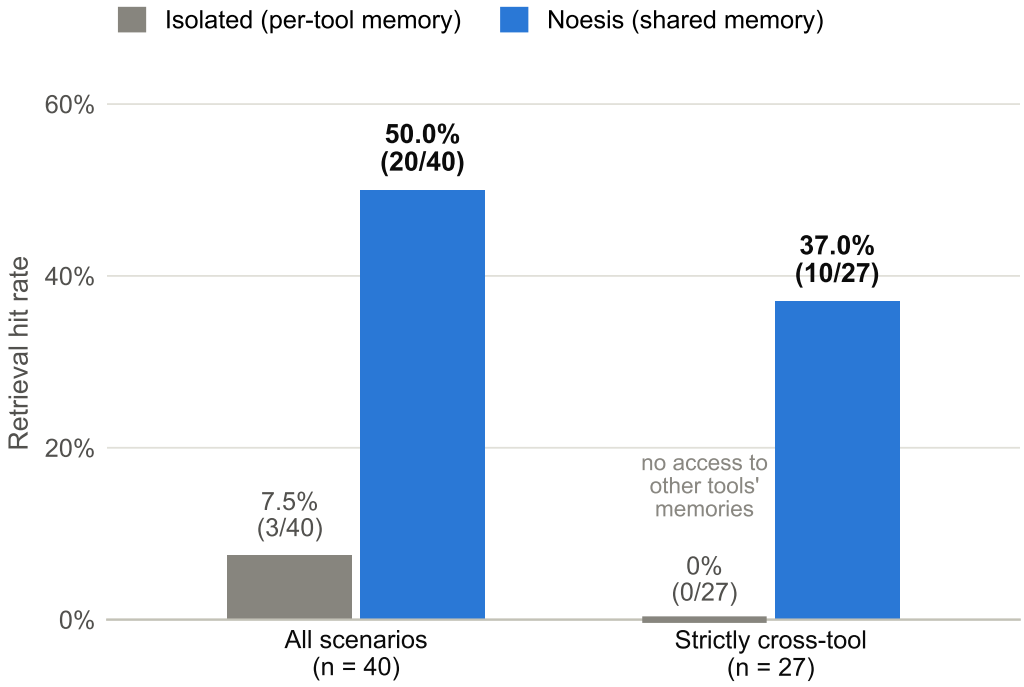


Figure 5: Cross-tool retrieval. Isolated per-tool memory scores 0% on strictly cross-tool queries by construction.

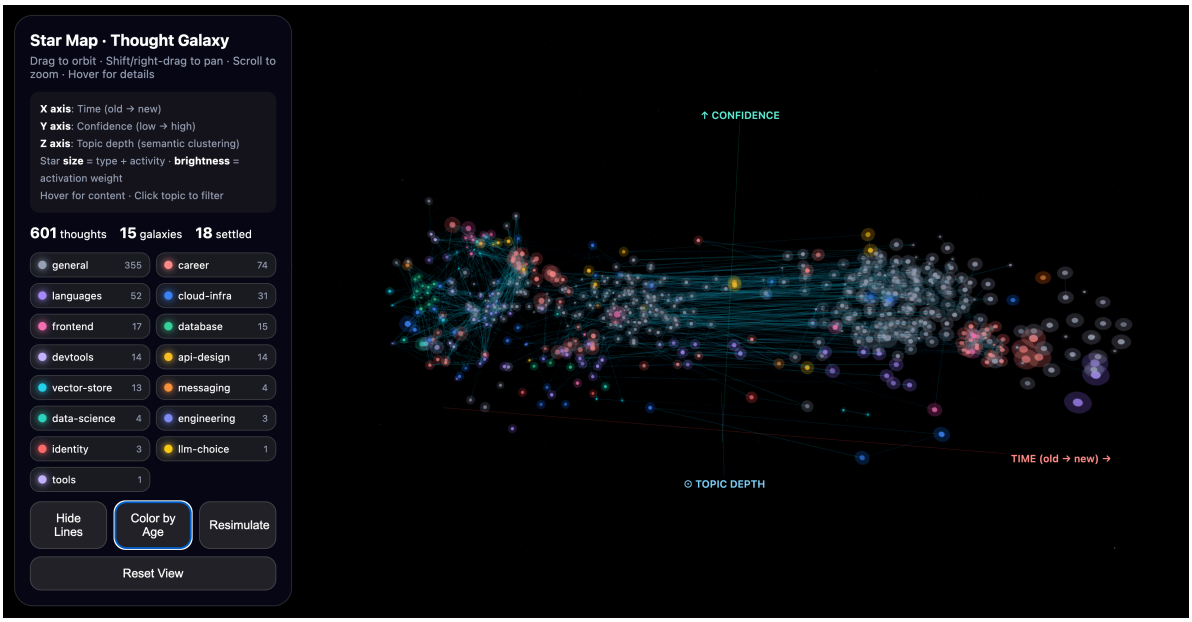


Figure 6: The star map. Semantic neighbors attract into volumetric constellations. Time runs from older (left) to newer (right), height reflects confidence, and color encodes topic; star size encodes type and activation weight, and brightness also tracks activation weight.

6 The Star Map Visualization

The stored memory graph is visualized as a galaxy (Figure 6). Each thought is a star placed by a short force simulation. Weak springs pull each star toward its creation time on the horizontal axis and its confidence on the vertical axis; *semantic attraction* pulls it toward its most similar neighbors (embedding cosine ≥ 0.45); and *repulsion* keeps stars from overlapping while pushing different topics apart. The springs are weak enough that attraction wins locally, so clusters form from embedding similarity while time and confidence stay readable at a glance; topic labels only decide which clusters repel. This makes the memory store inspectable rather than a black box: older thoughts sit to the left, settled ones sit high, and hovering a star shows its text and confidence. The store shown is the local development store, populated by the experiment and demo runs rather than by real use: of its 601 thoughts, 355 sit in the catch-all "general" topic, which the extractor assigns when no known topic matches, and 18 have settled. Even so, the view surfaces such patterns at a glance; here it shows how coarse topic assignment still is. The viewer is a local web tool shipped with the system; it opens the store read-only, so inspecting memory never changes it.

7 Discussion

When memory helps most. Gains concentrate on questions that are unanswerable without personal context (job, stack, stances). On mainstream questions both groups often answer correctly, which compresses the measured gap on those items. Other factors push the other way (see Limitations), so these results should not be read as a bound in either direction.

Memory can close the local–cloud gap. gemma3:4b posts the largest gain (+48.2pp; +139% over its baseline), approaching Gemini Flash-Lite once memory is injected. The benefit is not monotonic in model size: qwen2.5:3b gains least (+25.5pp), consistent with the comprehension ceiling in §5.1. Memory helps small local models most when they can use the injected context, which matters for privacy-sensitive and offline settings.

Retrieval precision is the current bottleneck. The cross-tool misses and several A/B misses reduce to ranking failures, not architecture failures: the memory is stored but ranks below the cut. Improving ranking (rerankers, query expansion) is the highest-leverage next step. The supersession fix's residual 7% is different: in both remaining leaks the detector never fired, once because "I now believe microservices are better" uses no listed replacement verb and names no known domain, and once because "I switched to Node.js" has the verb but Node.js is missing from the domain map. A broader mind-change detector is the matching fix.

Negative results. I report that the supersession check initially never ran, so stale stances leaked 93% of the time until it was fixed (§3.3); that core-fact and recency signals added no recall in my ablation suite; that a standard top-5 RAG matches Noesis's ranking at lower cost on the scale benchmark; and that a 3B model appears comprehension-capped. I consider these findings part of the contribution.

8 Future Work

Thought branching. When a new judgment deviates from history by more than a threshold ($\delta = 1 - \cos > 0.45$), Noesis should create a *branch* linked via `evolved_from` rather than superseding,

preserving the full trajectory of a stance. This enables metrics I cannot compute today: stabilization depth, branching rate, and post-branch divergence—a longitudinal record of how a user's positions evolve.

Structured injection. Replacing flat context lines with a structured `user_thought_graph` (anchor topic, trajectory with weights, cross-links) may improve the ability of small models to exploit injected memory, directly addressing the qwen2.5 comprehension ceiling.

Stronger evaluation. LLM-as-judge scoring to complement keyword hits; distractor questions; a second cloud model to replicate the Gemini result; longitudinal deployment data beyond synthetic profiles.

9 Limitations

Scoring is keyword-hit, which is objective but coarse; semantic correctness may differ in both directions. Personas are synthetic; build/test alignment may overstate real-world gains, and real conversations are noisier. The cloud A/B uses one provider (Gemini Flash-Lite) and n=110 rather than the planned 150: free-tier rate limits stopped the 30 June run after 111 questions. A later session added 35 more, but because the `gemini-flash-lite-latest` alias may have pointed to a different model version by then, I use only the first 110 questions (22 complete profiles) from the single June run; all significance thresholds are still exceeded by wide margins. Retrieval-layer experiments measure what reaches the prompt, not final answer quality. The supersession detector relies on replacement verbs and a domain keyword map; paraphrased mind-changes outside its lexicon will be missed. The supersession fix was developed and validated on the same 30 scenarios, so its 93% → 7% result is in-sample; held-out, paraphrased mind-changes are needed to measure it fairly. The scale experiment's naive-RAG baseline injects every stored memory; the top-5 baseline added in §5.3 is the stronger comparison, and on this benchmark it ties Noesis on ranking. The ablation and poisoning leak counts were corrected in October 2026: the original keyword-based metric also counted settled memories that merely shared a word with a tentative one (for example, "Kafka"). After an external review, I corrected the ablation further: it now scores exact memory identity (the old phrase matcher credited any memory sharing wording such as "I prefer"); it drops "Precision@5", which counted non-tentative results rather than relevant ones; it runs every variant through the same context builder (the old core-fact and recency rows could not differ from the full system); it applies the memories' ages, which the test store had silently dropped; and its recent-window and random baselines can now return tentative memories. Together these changes lowered the weak baselines (random 45% → 12%, w/o semantic 49% → 24% Recall@5) while Noesis moved from 100% to 99%.

Noesis also adds an attack surface. The local proxy sees every request, including provider API keys and full conversations; enabling cloud extraction sends conversation text to a third-party API, which weakens the local-first guarantee; and because a single firmly stated sentence already reaches provisional (§3.2), a planted or mistaken statement can enter future prompts. The confidence gate filters hedged statements but is not a defense against deliberate poisoning.

10 Conclusion

Noesis shows that *what* a memory system injects matters as much as *whether* it injects. A confidence lifecycle earns injection rights through repetition and consistency, and its gate keeps unverified and

planted statements out of the prompt where plain retrieval lets them in; retrieval keeps cost flat as memory grows (35 times cheaper than injecting everything at n=500, though a standard top-5 RAG is cheaper still); and a shared per-user store makes cross-tool consistency possible where per-tool architectures are structurally bound to fail. The result is a local-first layer that makes a 4B local model answer personal questions about as accurately as a small cloud model (82.7% vs. 85.5% for Gemini Flash-Lite; no significant difference on this set)—and that keeps the user's reasoning in the user's own vault.

Acknowledgments

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References

[1] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.11511.

[2] Prateek Chhikara, Dev Khant, Sparsh Aryan, Harshit Singh, Tanmay Yadav, Srijan Chawla, Nitesh Khare, Shiv Muthuraman S, Aman Gupta, Shantam Mehta, Taran Parekh, Priyansh Deshmukh, Devansh Dhamecha, Ashwini Kankrej, Kinar Kotecha, Srikar Revi, MR Bedmutha, Devendra Khanna, and Chintan Modi. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025. ECAI 2025.

[3] Thomas M. J. Fruchterman and Edward M. Reingold. Graph drawing by force-directed placement. *Software: Practice and Experience*, 21(11):1129–1164, 1991.

[4] Zheng Jiang, Frank F. Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7969–7992, 2023.

[5] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.

[6] Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[7] Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.

[8] Mem0. Introducing OpenMemory MCP. https://mem0.ai/blog/introducing-openmemory-mcp, 2025. Mem0 blog, May 13, 2025.

[9] Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Lin, Sarah Wooders, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

[10] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023.

[11] Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. Zep: A temporal knowledge graph architecture for agent memory. *arXiv preprint arXiv:2501.13956*, 2025.

[12] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008.

[13] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *International Conference on Learning Representations (ICLR)*, 2025.

[14] Wujiang Xu, Kai Mei, Hang Gao, Juntao Tan, Zujie Liang, and Yongfeng Zhang. A-MEM: Agentic memory for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[15] Liangwei Yang, Zhiwei Chen, Lixin Xia, Chen Yang, Yu Zhang, and Ben Kao. PALR: Personalization aware LLMs for recommendation. *arXiv preprint arXiv:2305.07622*, 2023.

[16] Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. MemoryBank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.